Fundamentals of Accounting & Finance

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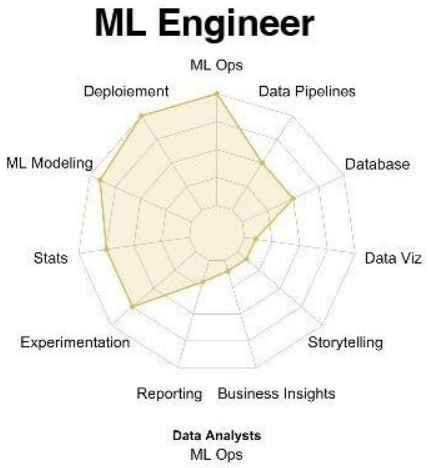
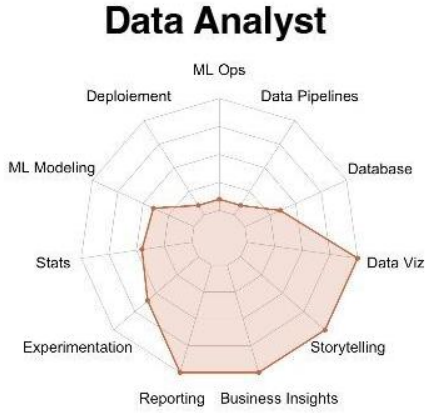
Where You Stand in the Data Analytics Spectrum?





The Accountant's Role

- The role of accounting:
  - To check transactions and operations, as they occur.
  - To record them in a consistent manner
  - To report the results in standardized form
- It is not the role of accounting:
  - Forecast the future, no matter how tempted.
  - Value assets or operations.
- **Bluntly put, an accountant is a historian, chronicling events that have already occurred, not predicting the future.**

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The Accounting Questions

- **What do you own as a business?**
  - List out the assets that a business has invested in, and how much it spent on those investments and perhaps what these assets are worth today.
- **What do you owe as a business?**
  - Specify the contractual commitments that a business must meet, to stay in business. Simply put, this should include all borrowings, but is not restricted to those only.
- **How much money did you make?**
  - Measure the profitability of the business, both with accounting judgments on expenses, and based upon cash in and cash out.
- **How much free cash-flow did you have?**
  - Measure the level of free cash-flow or the spending power the company has without relying on any other sources of capital.

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The Accounting Statements

- **The Balance Sheet**
  - It summarizes what a firm owns and owes at a point in time, as well as an estimate of what equity is worth (through accounting eyes).
- **The Income Statement**
  - It reports on how much a business earned in the period of analysis, while providing detail on revenues and expenses.
- **The Statement of Cash Flows**
  - It reports on cash inflows and outflows to the firm during the period of analysis and allows for a measure of cash earnings (as opposed to accounting earnings) and cash flows.

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The Accounting Statements: Balance Sheet

It summarizes what a firm owns and owes at a point in time, as well as an estimate of what equity is worth (through accounting eyes).

Assets		Liabilities	
Long Lived Physical Assets	**Fixed Assets**	**Current Liabilties**	Short term obligations
Short Lived Assets	**Current Assets**	**Debt**	Long term debt
Investments in Securities & other business	**Financial Assets**	**Other Liabilities**	Other long term obligations
Assets which are not physical	**Intangible Assets**	**Equity**	Shareholders' Equity

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The Accounting Statements: Income Statement

It reports on how much a business earned in the period of analysis, while providing detail on revenues and expenses.

	Item	*Explanation*
Start with	Revenues	Accountant's estimate of the revenues/sales generated by any transactions made the business during the period.
Net out	Cost of Goods Sold	Expenses associated with producing products or services that represent top line sales
To get	Gross Profit	Production profitability
Net out	Other Operating Expenses	Includes selling, general and administrative and other expenses associated with operations, but not directly tied to producing goods and services
To get	Operating Profit	Profitability of business
Net out	Financial Expenses	Expenses associated with the use of non-equity capital, especially debt.
To get	Taxable Income	Income for equity investors, prior to taxes
Net out	**Taxes**	Taxes due on taxable income
To get	**Net Income**	Income for equity investors, after taxes

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The Accounting Statements: Cash Flow Statement

It reports on cash inflows and outflows to the firm during the period of analysis and allows for a measure of cash earnings (as opposed to accounting earnings) and cash flows.

Net cash flow from operations, after taxes and interest expenses	Cash Flows From Operations
Includes divestiture and acquisition of real assets (capital expenditures), disposal and purchase of financial assets and cash spent on acquisition of other firms.	+ Cash Flows From Investing
Net cash flow from the issue and repurchase of equity, from the issue and repayment of debt and after dividend payments.	+ Cash Flows from Financing
Net cash flow, after investing and financing	= Net Change in Cash Balance

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The Accounting Standards

- **Accounting is a rule-driven process**, and over time, those rules have been formalized, especially for publicly traded companies.
- This formalization is driven by two considerations:
  - Standardization, to allow for comparisons across companies
  - First principles, to ensure that earnings, asset value and cash flows measure what they are supposed to measure.
- While accounting standards around the world remain different, they have converged (for the most part) around two standards:
  - **GAAP (Generally Accepted Accounting Principles),** representing rules developed by FASB (Financial Accounting Standards Board) to cover US financial reporting.
  - **IFRS (International Financial Reporting Standards)**, representing rules developed by IASB (International Accounting Standards Board) for companies listed globally, followed by about 90 countries as of 2020.

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The Accounting Standards: Accrual versus Cash Accounting

- In **accrual accounting**:
  - You record transactions when they occur, rather than when cash flows occur.
  - Revenues are recorded when a product or service is sold, not when the customer pays for that product or service.
  - Expenses are recorded consistently, with the expenses associated with producing the sold product or service shown in the period, even though you may have spent the money in a prior period or will not pay until a future period.
- In **cash accounting**:
  - You record revenues when you get paid for providing a product or service, and expenses when you pay.

- Unless you are a small or personal business, you will have to follow accrual accounting rules.

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The Accounting Standards: Classifying Expenses

- **Operating expenses** are expenses associated with the operations of the business. That includes not only the direct costs of producing the product or service the firm sells, but also, other expenses associated with production, including S, G & A expenses.

- **Financing expenses** are expenses associated with the use of non-equity financing. Most often, this takes the form of interest expenses on debt.

- **Capital expenses** are expenses that provide benefits over many years. For a manufacturing company, these can take the form of plant and equipment. For non-manufacturing companies, they can take on less conventional and tangible forms (and accounting has never been good at dealing with these).

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What is Finance?

- The raw material that we use to do financial analysis and valuation almost always takes the form of accounting statements. Consequently, it behooves us all to understand how accountants think (even if we disagree with them) in putting these statements together.

- Think of Finance as a game:
  - 2 Players: “Lender/Saver” & “Borrower/spender”
  - Borrower: has been exposed to some opportunities but lack the necessary funds to finance such opportunities.
  - Lender: has excess funds available

- Finance happens when the borrower borrows money from the lender and be able to take the opportunities, he thinks it worthwhile to take.
- When this happens, the borrower acquires capital needed to increase his “real assets” through selling “financial assets” to the lender.

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Real & Financial Assets

- Real assets:
  - Any tangible assets that generates a stream of goods or services in the future. For example, machinery, land, building, vehicle, robots, antiques, etc.

- Financial assets:
  - Intangible, usually paper claims by one person against another.
  - Seller of financial assets (i.e., the borrower) establishes liabilities which is an obligation.
  - Buyer of financial assets (i.e., the lender) buys some rights to entitle some claims of the seller's assets in the future.

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Example of Financial Assets

- Writer: Harry the Borrower
- Terms: I, Harry the Borrower, hereby swear to God that, on Dec 31, 2024, I will pay 10 bucks (USD) to whoever holds this paper contract with my big name signed on it. Sue me if I don't fill my obligation.

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Payoff from a Game of Finance

		Borrower's actions	
		Pay Fully	Default
Lender's actions	Lend \$X at r	((1+r)\$X, (R-r)\$X)	(-\$X,\$X(1+R))
	Don't lend at r	((1+R_0)\$X, 0)	((1+R_0)\$X,0)

- R = the rate of interest the borrower can earn from exploiting his opportunities.
- R_0 = the rate of interest the lender can earn from exploiting his own opportunities.
- r = the rate of interest embedded in the financial contract.

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Payoff from a Game of Finance

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- R = the rate of interest the borrower can earn from exploiting his opportunities.
- R_0 = the rate of interest the lender can earn from exploiting his own opportunities.
- r = the rate of interest embedded in the financial contract.
- If finance happens:
  - Lender's action = Lend \$X at r. Social surplus increases if R> R_0
  - The more interesting question is: "why in some situations the lender may not always take such socially optimal action?"

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Conditions for Finance

- The required conditions needed for ‘financial transaction’ to happen:
  - When the 2 players can really match up with each other. (The borrower successfully meets his potential lender.) (MATCHING)
  - When they can successfully negotiate a finance contract. (NEGOTIATION)

- Question:
  - We know financing is good as it generates gross surplus (i.e., $R > R_0$). But what are the costs involved in such financing activity? If cost > gross surplus, net surplus is negative.

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Costs and Benefits

- Cost of Financing
  - Search costs
  - Transaction costs
  - Negotiation costs (trust?)
  - RISK!!! Cost of assessing risk and monitoring risk.

- If the presence of financial market can overcome these costs, the benefits of finance would be:
  - Facilitate the channeling funds from saver to borrower in more efficient ways.
  - Savers are better off by lending and earn higher interest. Borrower can invest in productive opportunities. Overall, the society is better off.
  - Theoretically, if costs of financing is low, more of such financing activities will boost the economy.

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Financial Intermediaries

- Financial market has players (usually big ones) like the banks, insurance companies, etc, that act as financial intermediaries, and also primary and secondary markets.

- Primary market – trading of new issues of securities such as bond or stock.
- Secondary market – trading of securities that have been previously issued. This injects "liquidity" to the primary market, leading to lower risk for both lenders and borrowers.

- It is set up to overcome/lower the costs as mentioned.
- Lower transaction costs, equilibrium market risk-premium dictates the price (save bargaining costs), lower search costs, established trust, provide risk monitoring more effectively.

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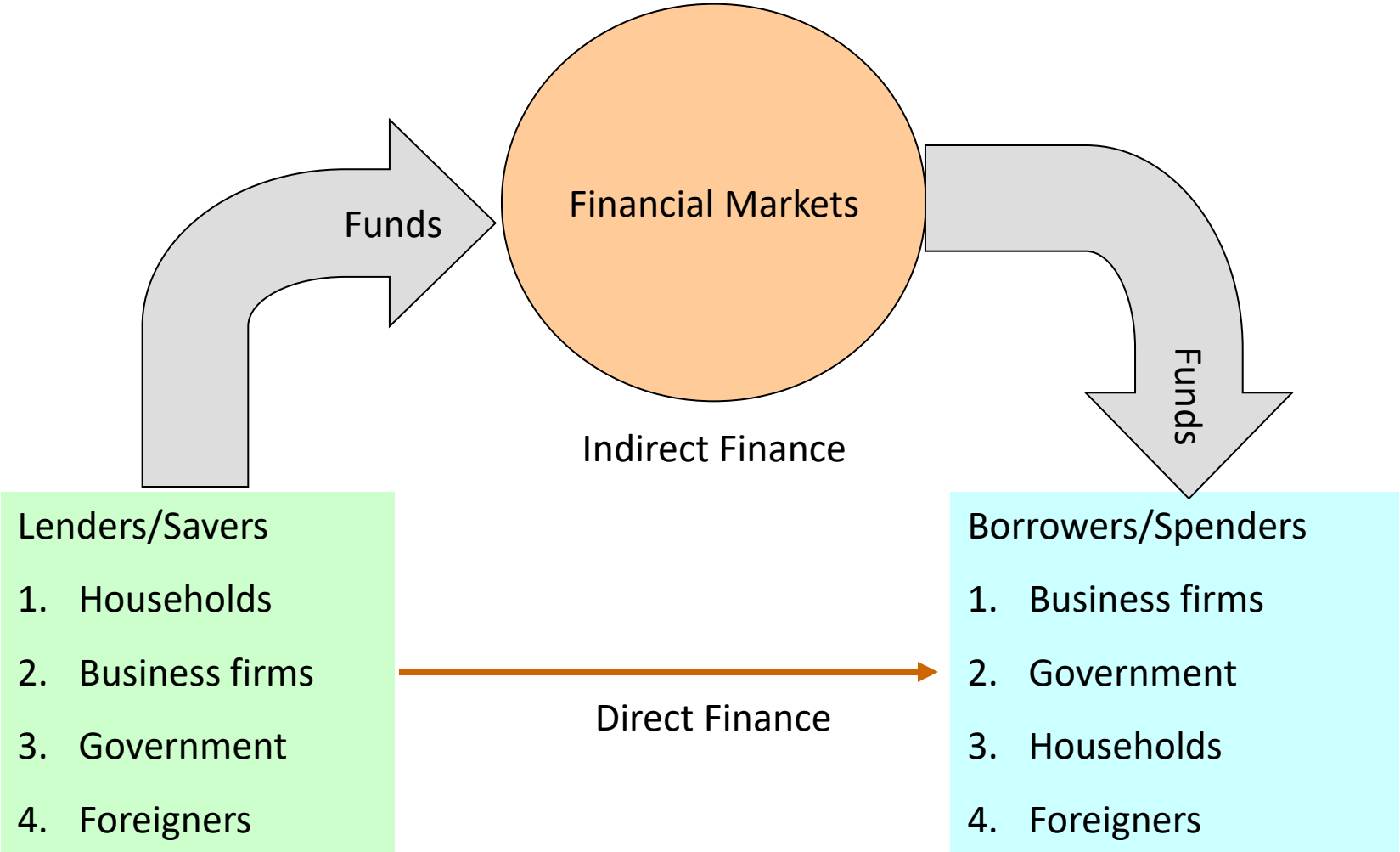
Flows of Funds



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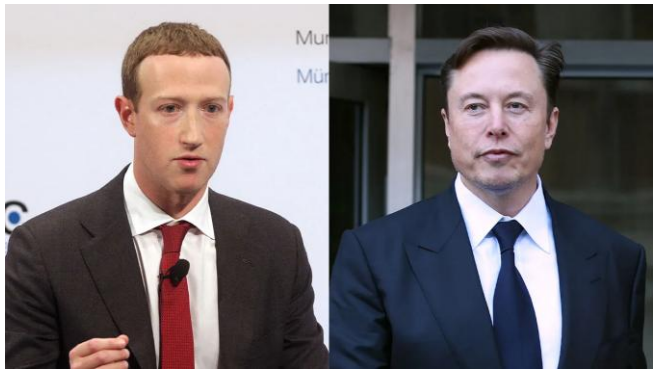
Summary of Finance

- In short, finance is an activity that involves borrowers and lenders. They can finance directly, or they can finance through intermediaries.
- It is easy if you lend money to your girlfriend, but if you want to lend money to these **2 guys**, you can also do so through participating in the financial market. No matter how much you admire these two guys and how confident you are to their production, you may never run into them throughout your entire life. So essentially, financial market provides a more efficient way of channeling funds.
- If the financial market functions well, it should be transparent. People would feel comfortable dealing with it. Default rates should be low. Important banks should have low bad debt. And transaction costs would be low, and most important of all, funds will be channeled more efficiently. The society, as a whole, will be better off with a healthy financial market.



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What is Business Finance?

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What is Corporate (Business) Finance?

- Wall Street Journal Defined it: “The Business of Financing Business”
- Just as Real Estate Finance deals with financing of an assets and Consumer Finance deals with the financing of an individual, Corporate Finance deals with the financing of corporation

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Traditional Accounting View of the Firm

The Balance Sheet

Assets		Liabilities	
Long Lived Real Assets	Fixed Assets	Current Liabilties	Short-term liabilities of the firm
Short-lived Assets	Current Assets	Debt	Debt obligations of firm
Investments in securities & assets of other firms	Financial Investments	Other Liabilities	Other long-term obligations
Assets which are not physical, like patents & trademarks	Intangible Assets	Equity	Equity investment in firm

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The Financial View of the Firm

Assets		Liabilities	
Existing Investments Generate cashflows today Includes long lived (fixed) and short-lived(working capital) assets	Assets in Place	Debt	Fixed Claim on cash flows Little or No role in management *Fixed Maturity* *Tax Deductible*
Expected Value that will be created by future investments	Growth Assets	Equity	Residual Claim on cash flows Significant Role in management *Perpetual Lives*

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Questions in Business Finance

- Capital Investment Decision:
  - What long-term investments should the firm engage in?
- Capital Structure Decision:
  - How can the firm raise the money for the required investments?
- Payout Decision:
  - What to payback to shareholders?
- Working Capital Management Decision:
  - How much short-term cash flow does a company need to pay its bills?
- Risk Management Decision:
  - What hedging policy is right for the firm?

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The Capital Investment Decision

The Balance-Sheet Model of the Firm

Current Assets

- Fixed Assets:
  - Tangible
  - Intangible

What long-term investments should the firm engage in?

Current Liabilities

Long-Term Debt

Shareholders' Equity

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The Capital Investment Decision - Example

Project 1: Cash flows of the project

Date	0	1	2	3
Cash Flows	$-100	$5	$5	$80

Project 2: Cash flows of the project

Date	0	1	2	3	4	5
Cash Flows	$-100	$5	$5	$5	$5	$110

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The Capital Structure Decision

The Balance-Sheet Model of the Firm

Current Assets

- Fixed Assets:
  - Tangible
  - Intangible

How can the firm raise the money for the required investments?

Current Liabilities

Long-Term Debt

Shareholders' Equity

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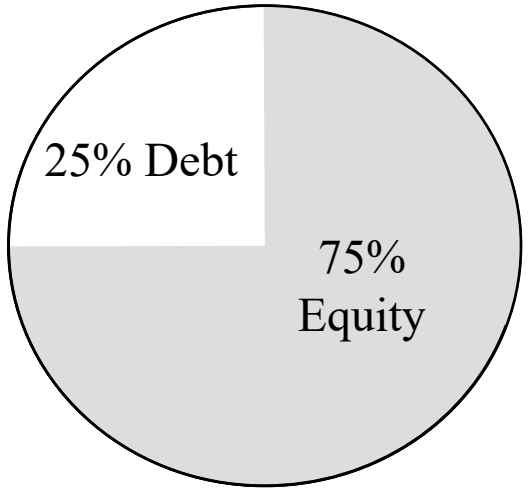
Capital Structure

The value of the firm can be thought of as a pie.

The goal of the manager is to increase the size of the pie.

The capital structure decision can be viewed as how best to slice up the pie.



If how you slice the pie affects the size of the pie, then the capital structure decision matters.

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Capital Structure

- Firm value = value of debt (bonds) + value of equity (stocks)
  - Bonds = loan agreements
  - Stocks = share certificates

- A debt is a promise by the borrowing firm to repay a fixed dollar amount by a certain date.

- The shareholder's claim on firm value is the residual amount that remains after the debtholders are paid.

- If the value of the firm is less than the amount promised to the debtholders, the shareholders get nothing.

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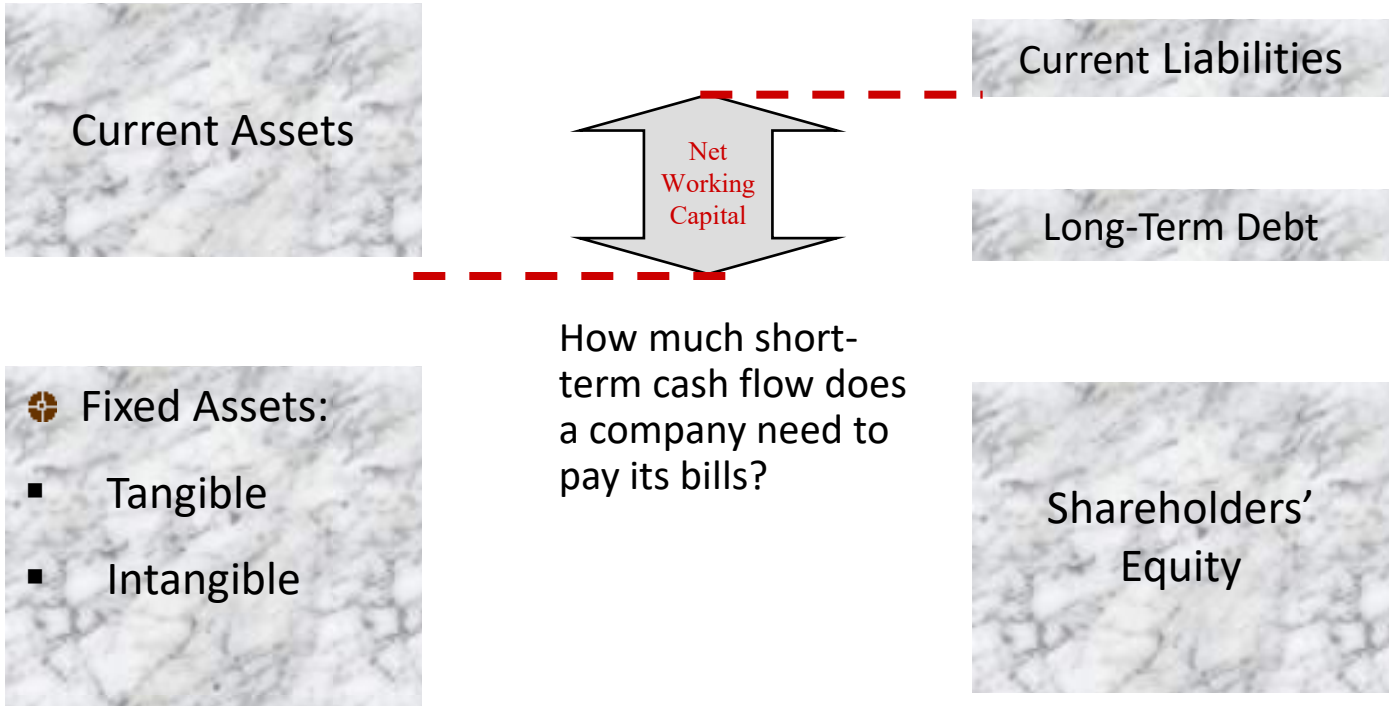
Working Capital Decision

The Balance-Sheet Model of the Firm



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Payout Decision

- Many companies pay a **regular cash dividend**
  - Public companies often pay quarterly
  - Sometimes firms will throw in an extra cash dividend
  - The extreme case would be a liquidating dividend

- Often companies will declare **stock dividends**
  - No cash leaves the firm
  - The firm increases the number of shares outstanding

- Some companies declare a **dividend in kind**
  - Wrigley's Gum sends around a box of chewing gum
  - Dundee Crematoria offers shareholders discounted cremations

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Risk Management Decision

- Create Firm Value through:
  - Smart Investment Decisions.
  - Smart Financing Decisions.
  - Smart payout policy decision
  - Smart Working Capital Management Decision

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Why Study Business Finance?

- It is important to understand financial view of the corporate firm.
- Firm exits to solve transaction cost associated with the price mechanism. It is a nexus of legal contract that joins physical assets and human capital to pursue value creation that cannot be easily replicated by others.
- The corporate form of business is the standard method for solving the problems encountered in raising large amounts of cash.
- However, businesses can take other forms.

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Forms of Business Organization

- The Sole Proprietorship
- The Partnership
  - General Partnership
  - Limited Partnership
- The Corporation
- Trust
- Difference lies in:
  - Liquidity and Marketability of Ownership
  - Control
  - Liability
  - Continuity of Existence
  - Tax Considerations

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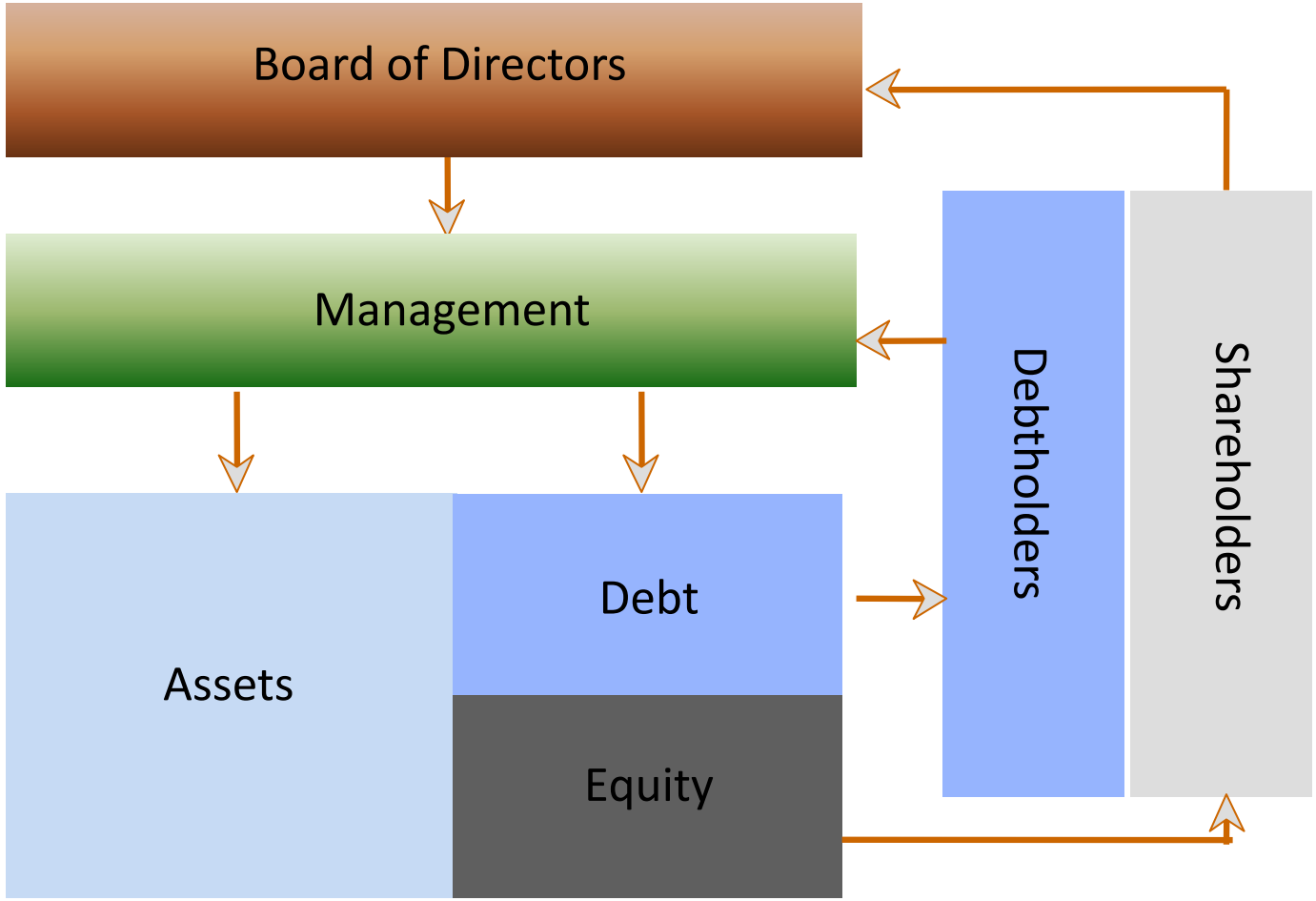
Corporate Firm: Separation of Ownership and Control



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The Goal of the Corporation

- Need an objective by which management can make decisions.

- Decision rules must be consistent with the objective:
  - Maximize profits?
  - Minimize costs?
  - Minimize risk?
  - Maximize sales or market share?
  - Maintain steady earnings growth?
  - Survive in business?

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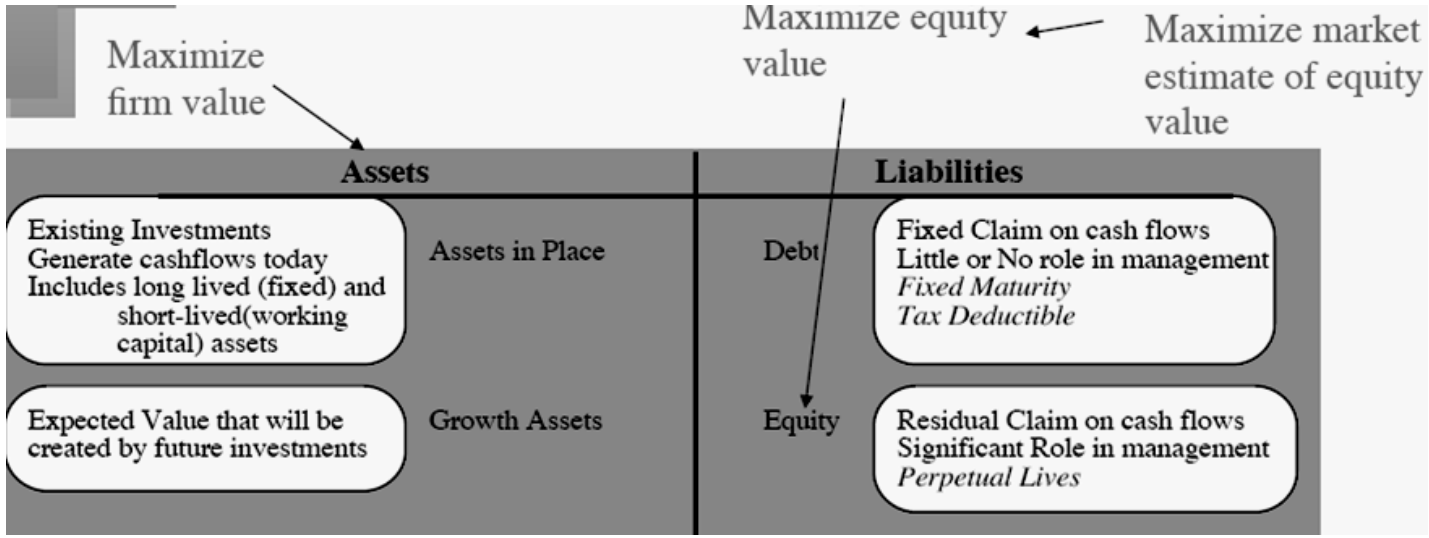
The Goal of the Corporation

- In corporate finance, the objective is to maximize the value of the firm.
  - Maximization of the wealth to all investors (stock and bond).
- A narrower objective is to maximize shareholder wealth.
  - When the stock is traded and markets are viewed to be efficient, the objective is to maximize stock price.
- Under certain conditions, these objectives are equivalent.



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Corporate Governance

- To determine whether management's actions are consistent with this objective, we need to understand the corporate governance structure and practices of the firm.

- "Corporate governance is the system by which business corporations are directed and controlled. The corporate governance structure specifies the distribution of rights and responsibilities among different participants in the corporation, such as, the board, managers, shareholders and other stakeholders, and spells out the rules and procedures for making decisions on corporate affairs. By doing this, it also provides the structure through which the company objectives are set, and the means of attaining those objectives and monitoring performance," **OECD** April 1999.

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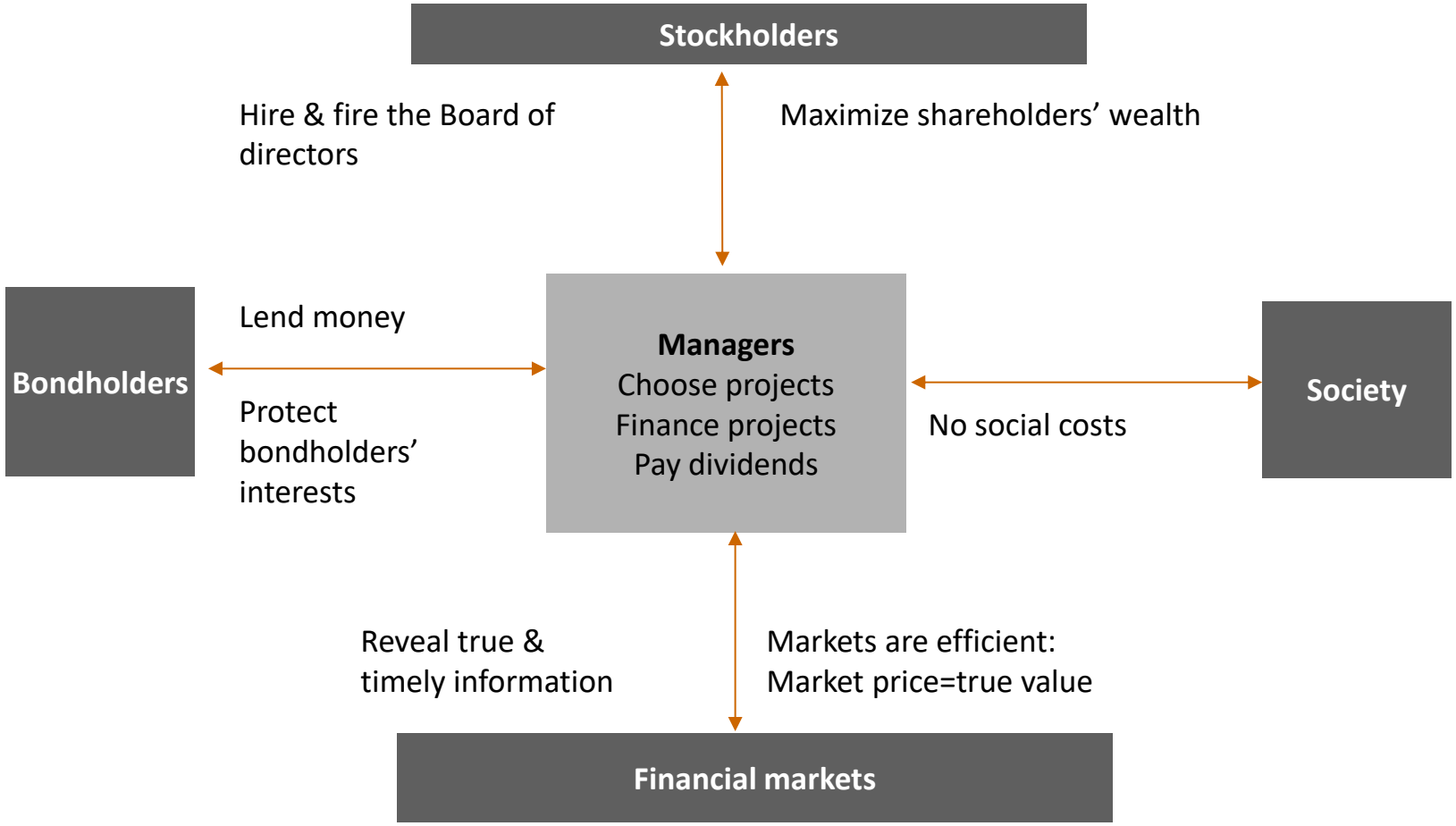
Governance Structure of a Corporation



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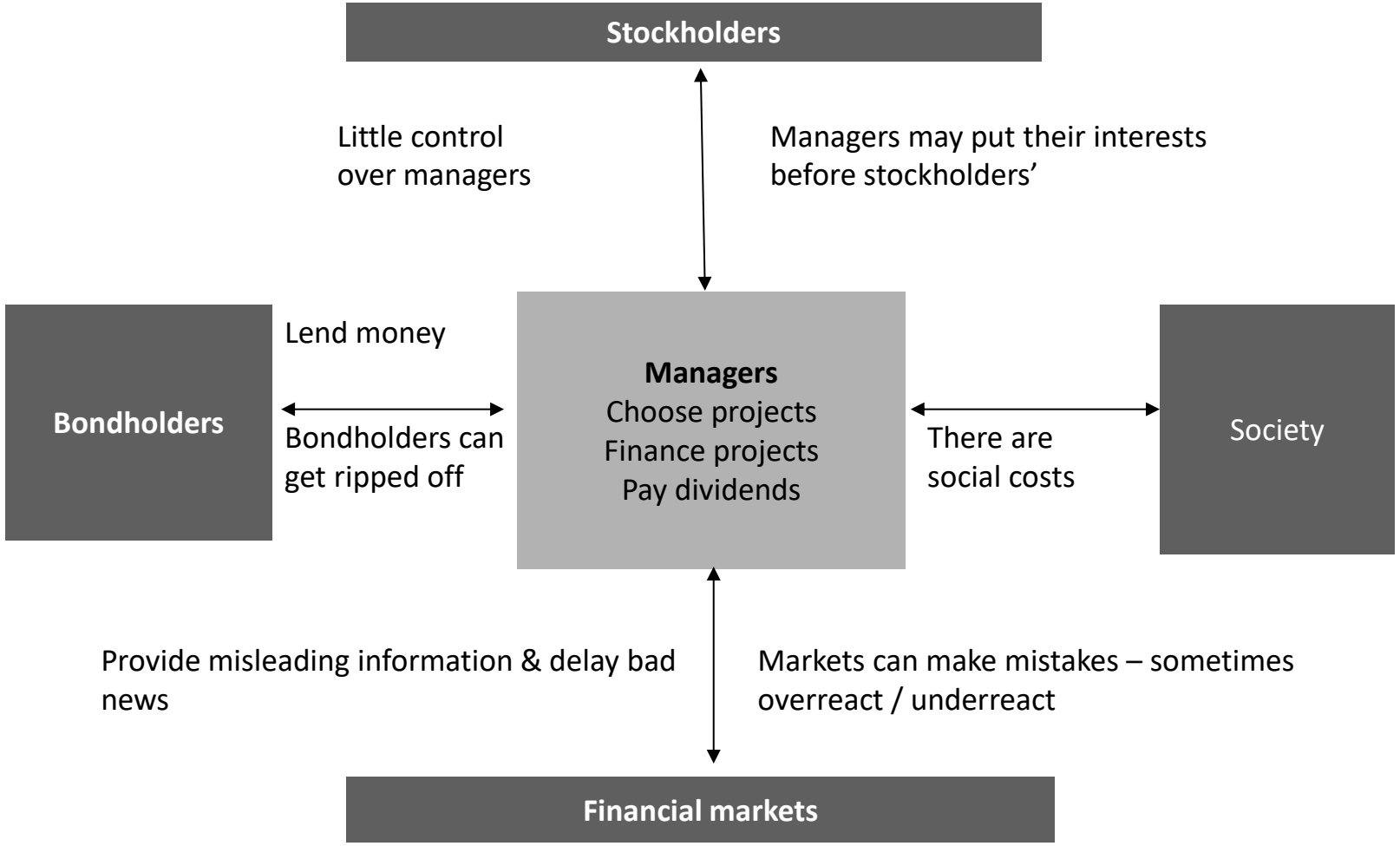
But, in Reality...



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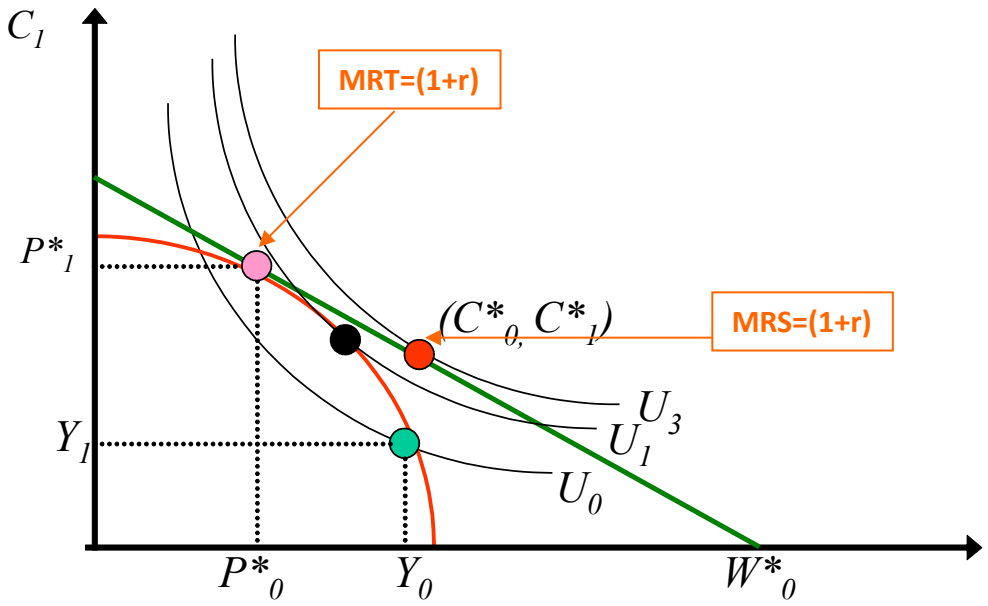
Fisher Separation Theorem

- The decisions of production and consumption involve 2 distinct steps:
  - **<u>1st step:</u>** Choosing production point by moving along POS and produce at the point where **MRT = (1+ r)**, this means return on the marginal investment is just equal to market interest rate.
  - **<u>2nd step:</u>** Choosing consumption point by moving along the capital market line and consume at the point where **MRS = (1 + r)**, this means subjective rate of time preference is equal to market interest rate.
- We call this the "FISHER SEPARATION THEOREM". The important point is that the production point is governed solely by objective criteria, namely, the set of opportunities available and the market interest rate. This is independent of individual's subjective rate of time preferences.



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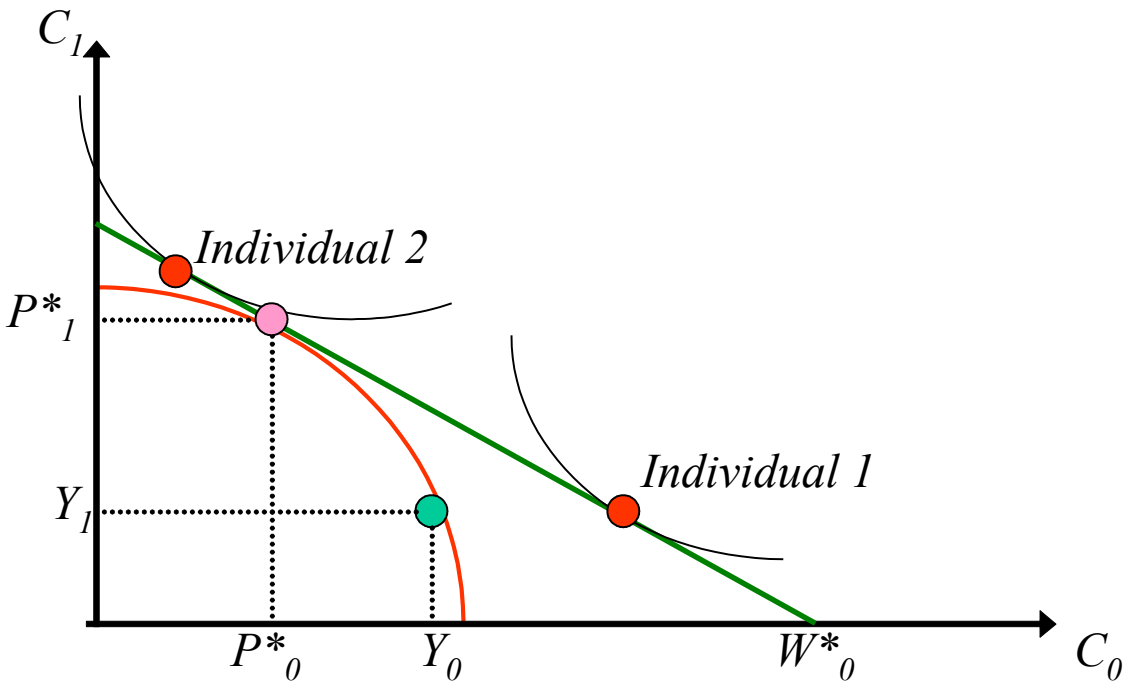
Fisher Separation Theorem

- **Implication 1:** As the graph below shows, with two different individuals that differs only in their subjective preferences, given the same opportunity set, both of them would choose the exact same point of production regardless of the difference of their preferences.



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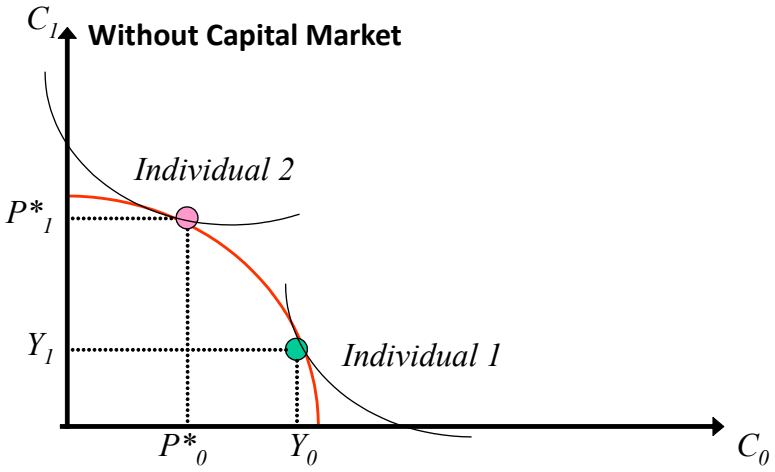
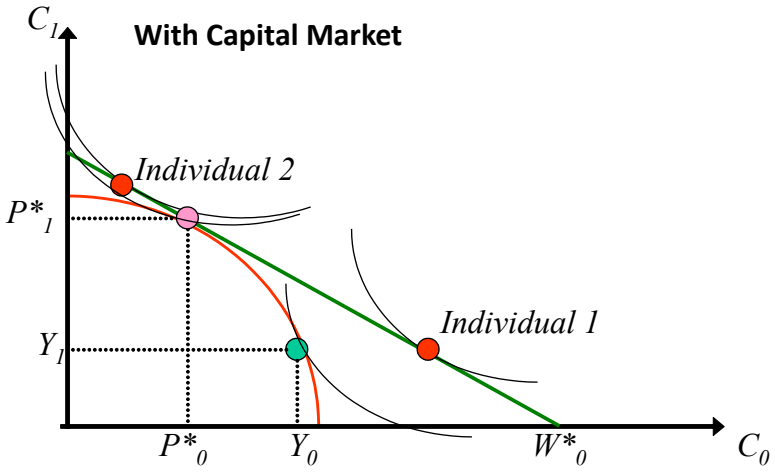
Fisher Separation Theorem

- **Implication 2:** The capital market channel funds from the lenders to the borrowers. Setting demand = supply, we have a market-determined r. Given the individual's exposure to his own production opportunities, he may decide whether to lend or borrow money. By allowing lending and borrowing, those who need money can get financed, while those have excess fund will be able to lend out and earn interest. Everyone is made better off.
- In short, as shown below, capital market is important because everyone can be happier with it.
- The sources of happiness enhancement: 1) measurement: capital market gives a precise benchmark for what to invest and what not to invest. 2) financing: individuals can actually borrow and lend to invest in what ought to be invest.





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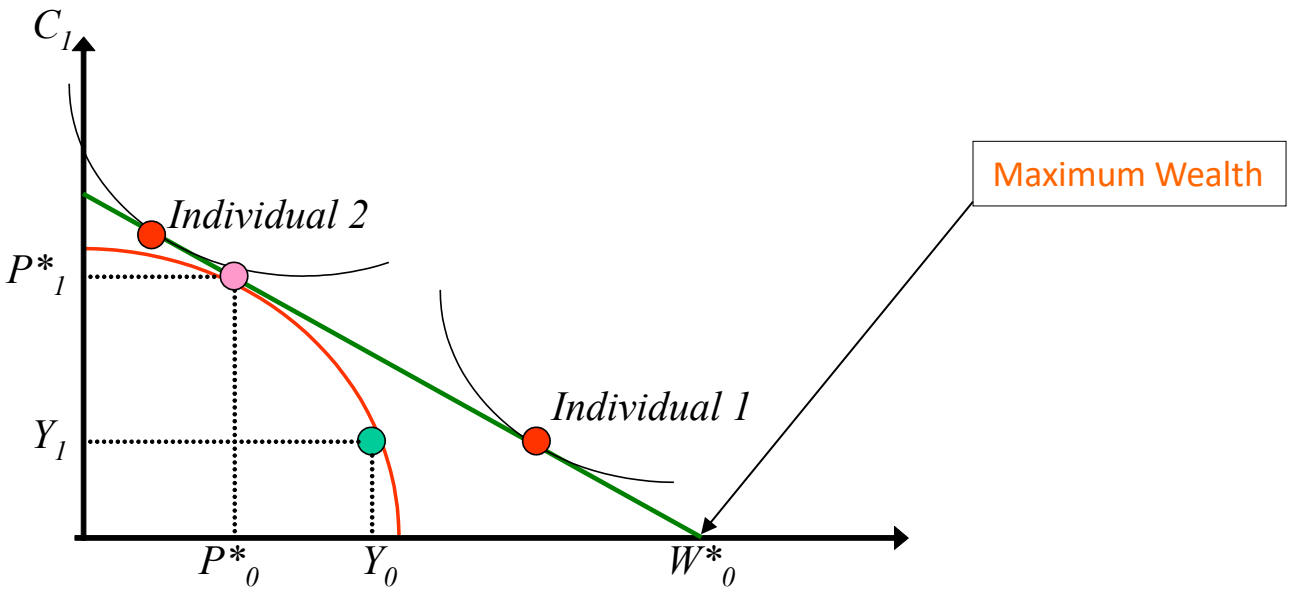
Fisher Separation Theorem

- **Implication 3:** Consider the following two investors investing all their money in the stocks of a single firm. Their well-being is thus tied to the well-being of the firm. Consider the firm is making decision of what to produce.
- Fisher Separation Theorem implies even the two investors differ in their subjective perception of how to consume between now and future, they both has one unified objective, i.e, to maximize their current wealth.
- Doing so means the firm can maximize its value. This is the same as investing until the return on the marginal investment is just equal to the cost of capital, i.e, the market interest rate. And the firm knows that its shareholders will unanimously agree on what it does.



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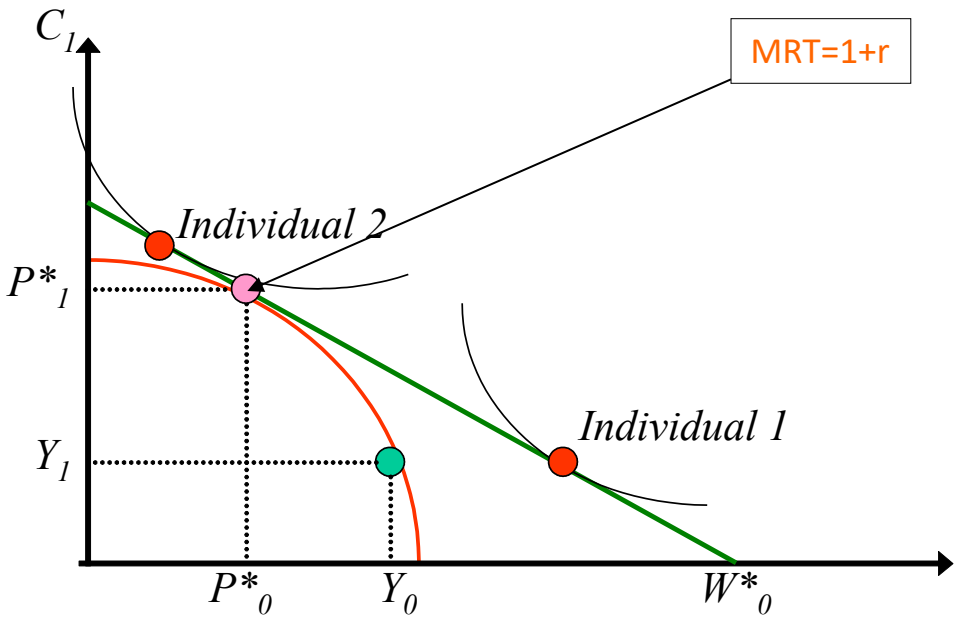
Fisher Separation Theorem

- **Implication 3:** MRT = (1 + r) is the point where both of the two individuals would agree for the firm to produce.
  - This is exactly the famous project selection rule, the "positive Net Present Value rule". The firm value is maximized by taking all projects that have positive NPV.
  - NPV = -initial investment + present value of future payout discounted by cost of capital.
  - Cost of capital = r



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How to Max Shareholder's Wealth?

- We again use **Fisher Separation Theorem**:
  - Given perfect and complete capital markets, the owners of the firm (shareholders) will unanimously support the acceptance of all projects until the least favourable project has return the same as the cost of capital.
  - In the presence of capital markets, the cost of capital is the market interest rate.
  - The project selection rule, i.e., equate marginal rate of return of investment = cost of capital (market interest rate)
  - Is exactly the same as the positive net present value rule:
- **Net Present Value Rule:**
  - Calculate the NPV for all available (independent) projects. Those with positive NPV are taken.
  - <u>At the optimum:</u> NPV of the least favourable project ~= zero
  - This is a rule of selecting projects of a firm that no matter how individual investors of that firm differ in their own opinion (preferences), such rule is still what they are willing to direct the manager to follow.

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First Principles of Business Finance

- Invest in projects that yield a return greater than the minimum acceptable hurdle rate with adjustments for project riskiness. On other words, invest in projects with the highest positive NPV.
- Choose a financing mix that minimizes the hurdle rate.
- If there are not enough investments that earn the hurdle rate, return the cash to stockholders.

- These decision criteria will be consistent with the objective of the firm: **Maximize the Value of the Firm.**

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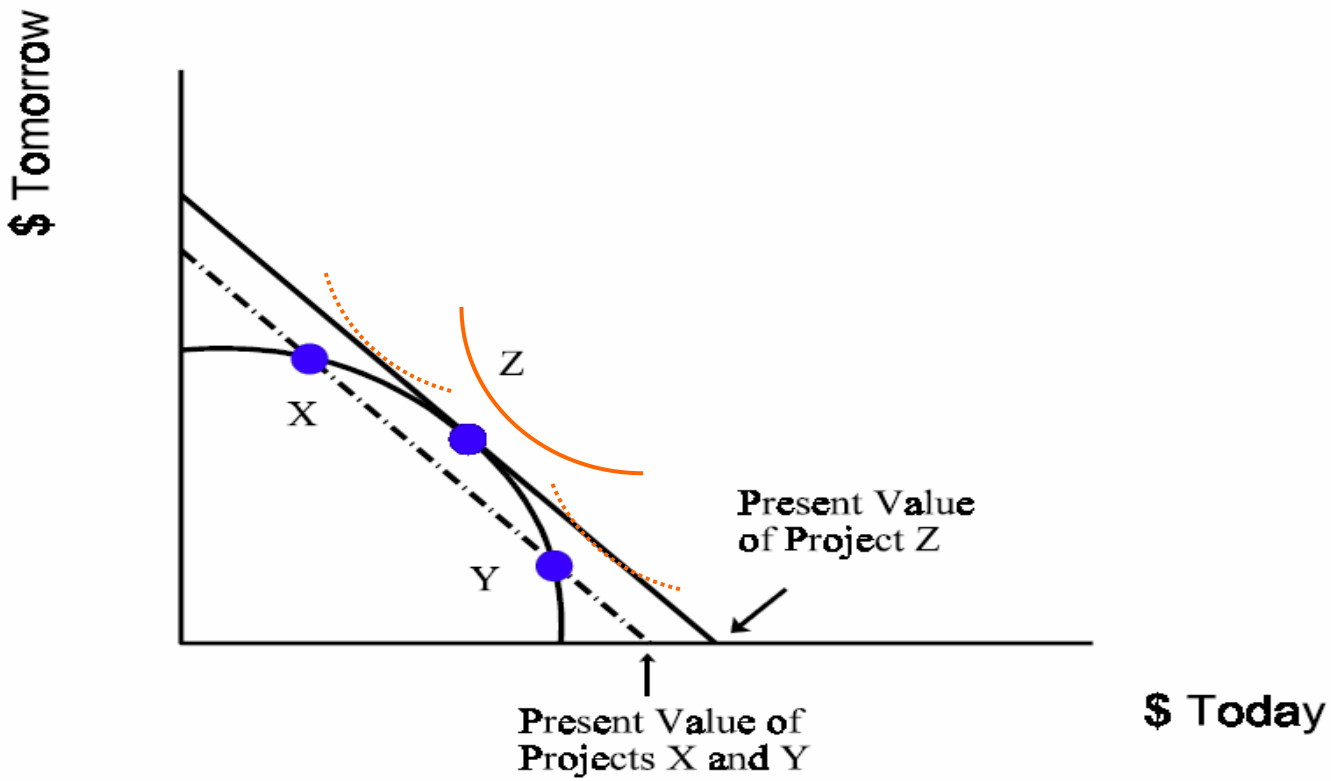
NPV Intuition



The graph displays three typical projects that are available to the firm on the investment opportunity curve. If either project X or Y is selected, then the investor can consume along the dashed line by accessing the capital markets. If project Z is selected then consumption can take place along the solid line. Project Z is the optimal project. The principle that *investors want the project with the highest PV* continues to hold.

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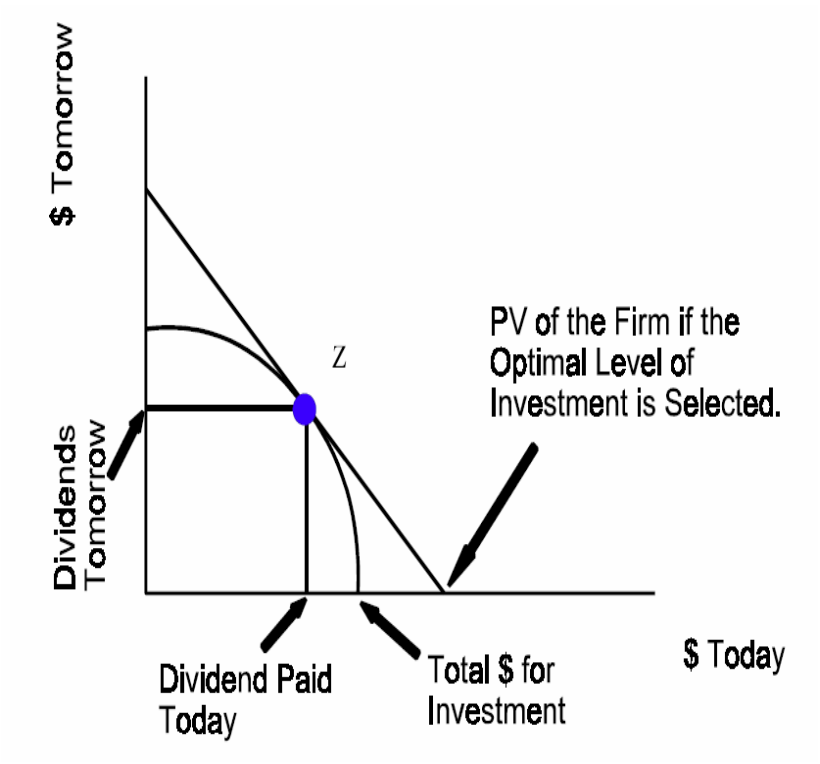
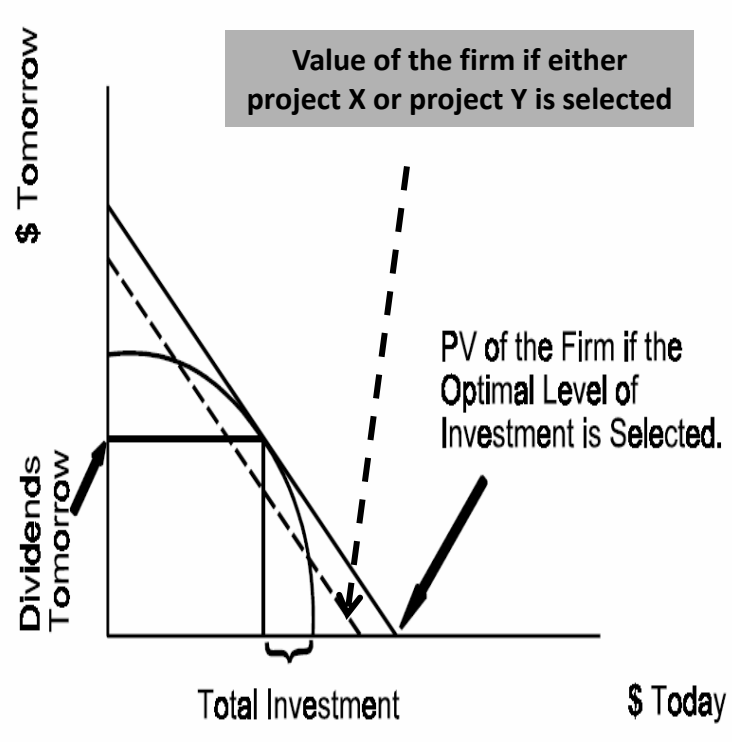
NPV Intuition





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Again and Again, Fisher Separation Theorem

- The separation principle implies that the maximization of the shareholder's wealth is identical to maximizing the present value of lifetime consumption
- Since borrowing and lending take place at the same rate of interest, then the individual's production optimum is independent of his resources and tastes
- If asked to vote on their preferred production decisions at a shareholder's meeting, different shareholders will be unanimous in their decision
  - **Unanimity principle**
- Managers of the firm, as agents for shareholders, need not worry about making decisions that reconcile differences in opinion among shareholders i.e there is unanimity
- The rule is therefore
  - **Take projects until the marginal rate of return equals the market interest rate = taking all projects with +ve NPV**

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10 Principles of Finance

- **Time value of money:** A dollar today is worth more than a dollar tomorrow because money can earn interest over time.
- **Law of one price:** Identical assets traded in competitive markets should have the same price everywhere.
- **No free lunch:** Every financial opportunity has a cost; unusually high returns always come with trade-offs or risks.
- **Models are simplifications:** Financial models are useful but incomplete representations of reality and must be judged by their purpose and context.
- **Market prices matter:** An asset's market price is often the best estimate of its value because it aggregates available information.
- **Risk is central:** Risk is unavoidable in finance and must be explicitly measured and managed in all financial decisions.
- **Risk–return trade-off:** Higher expected returns require bearing higher risk; investors must be compensated for taking risk.
- **Flexibility has value:** Financial flexibility, such as that embedded in options, is more valuable when uncertainty is high.
- **Transparency and trust matter:** Well-functioning financial systems rely on transparency, verification, and trust among participants.
- **Financial systems are universal, institutions are not:** The core functions of finance are universal, but the institutions that perform them vary across countries and over time.

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